

Firmware

- Lowest level of software that runs on a computer
- Begins the boot process and configures hardware devices
- Lets you enable/disable hardware
- Once Linux boots, it uses its own drivers to access hardware

BIOS/EFI/UEFI

- The most important firmware is installed on the computers **motherboard**
- In the past most computers used firmware called *Basic Input/Output System* (**BIOS**)
Since 2011, a new type of firmware known as *Extensible Firmware Interface* (**EFI**) or *Unified EFI* (**UEFI**) has become standard
- **Most manufacturers refer to EFI as BIOS**

Flash Memory

Firmware is stored in electronically erasable programmable read-only memory (**EEPROM**) - **flash memory**

IRQ

- Interrupt **ReQ**uest
- **signal sent to the CPU to handle an event**
e.g. keyboard input

IRQs are numbered 0 to 15, some modern computers have more

- some are reserved for specific purposes e.g. keyboard
- some are shared with common uses
- some are left for extra devices that may be added to the system

Common IRQ

1	Keyboard
12	Mouse

Examining IRQ

`cat /proc/interrupts`

```
james@laptop:~$ cat /proc/interrupts
           CPU0      CPU1      CPU2      CPU3      CPU4      CPU5      CPU11
0:         72         0         0         0         0         0         0  IR-IO-APIC  2-edge  timer
1:         77         0         0         0         15        0         0  IR-IO-APIC  1-edge  i8042
8:          0         0         0         0         0         0         0  IR-IO-APIC  8-edge  rtc0
9:       9092      25261         0         0         0         0         0  IR-IO-APIC  9-fasteoi acpi
14:          0         0         0         0         0         0         0  IR-IO-APIC 14-fasteoi INT3450:00
16:          0         0         0         0         10        0         0  IR-IO-APIC 16-fasteoi idma64.0, i2c designware.0, i801_smbus
17:       1959         0         0       1109         0         0         0  IR-IO-APIC 17-fasteoi idma64.1, i2c designware.1, snd_hda_intel:card
```

shows:

1. IRQ
2. number IRQs for a CPU
3. driver name

I/O Addresses

also called I/O ports

for CPU communication - maps location memory : hardware

usually mapped to a specific device

to view:

```
james@laptop:~$ sudo cat /proc/ioprocs
0000-0cf7 : PCI Bus 0000:00
0000-001f : dma1
0020-0021 : pic1
0040-0043 : timer0
0050-0053 : timer1
0060-0060 : keyboard
0064-0064 : keyboard
```

DMA

Direct Memory Address

alternative to I/O ports

Bypass CPU

- instead of having CPU transfer data between memory and a device
- *allows transfer directly between memory and device*

Channels

Pathway to allow transfer between memory and device

DMA channels in use:

```
james@laptop:~$ cat /proc/dma
4: cascade
```

channel 4 is being used

Hotplug and Coldplug Devices

Hotplug: can be attached/detached when **computer on** (hot)

Coldplug: only attached/detached when **computer off** (cold)

Coldplug

CPU, memory, PCI cards, SSDs

Hotplug

Rely on software to detect changes to the system as they're attached/detached

Hotplug utilities

utilities to manage hotplug devices

enable programs to learn about hardware

receive notifications when hardware configuration changes

1. sysfs

- a virtual filesystem
- mounted at /sys
- exports info about devices so user-space utilities can access the information

user space program	runs as an ordinary program run by user or as root
kernel space code	runs as part of the kernel can communicate with hardware helps hotplug devices access hardware

- **HAL Daemon**

Hardware Abstraction Layer Daemon

- a user-space program that runs all the time
- provides other user-space programs with hardware info

- **D-Bus**

Desktop **Bus** daemon (always running in the background)

- lets process communicate
- lets process register to be notified of events by other processes or hardware
e.g. new USB device plugged in

- **udev**

- virtual filesystem mounted at `/dev`
- creates device files as drivers are loaded/unloaded